**IN THE UNITED STATES PATENT AND TRADEMARK OFFICE**

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| **In re Patent Application of:** | Reinaldo Javier Aguiar Marcano |
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| **Examiner:** | Andrea Bennett |
| **Title of Invention:** | MEMORY SYSTEM FOR THE STORAGE AND RETRIEVAL OF PARAMETRIC AND SYMBOLIC DATA REPRESENTATIONS |

## REPLACEMENT ABSTRACT (37 CFR § 1.72(b))
## (COMMENCING ON A SEPARATE SHEET)

### ABSTRACT

A memory system and architecture designed for the storage and retrieval of data as parametric or symbolic representations rather than as fixed-width, evaluated numerical values. The system comprises a memory controller and a storage medium (such as DRAM, flash, or ROM) where data is stored as variable-length structures representing mathematical 'recipes' or expressions. The memory controller manages the dynamic allocation of physical memory to accommodate these structures and interfaces with a parametric data bus. This architecture ensures that the precision of data is not lost when it is moved from a parametric CPU to memory, enabling the persistence of perfectly represented fractional, irrational, and large-magnitude numbers throughout a complete computing system.